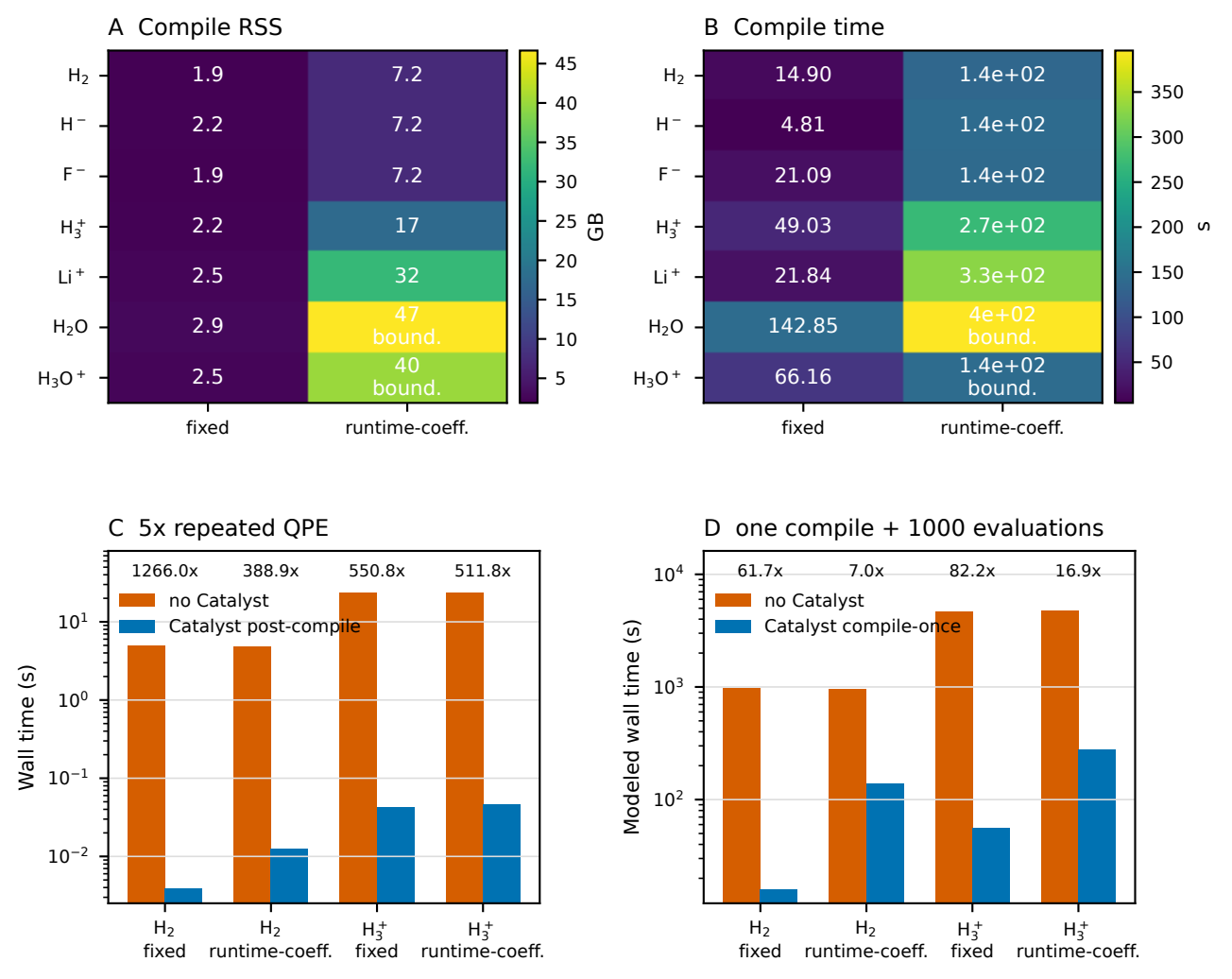
A Compile RSS
H2
H−
F−
H3+
Li+
H2O
H3O+
1.9
7.2
2.2
7.2
1.9
7.2
2.2
17
2.5
32
2.9
47
bound.
2.5
40
bound.
fixed
runtime-coeff.
45
40
35
30
25
20
15
10
5
GB
B Compile time
14.90
1.4e+02
4.81
1.4e+02
21.09
1.4e+02
49.03
2.7e+02
21.84
3.3e+02
142.85
4e+02
bound.
66.16
1.4e+02
bound.
350
300
250
200
150
100
50
s
C 5x repeated QPE
1266.0x
388.9x
550.8x
511.8x
no Catalyst
Catalyst post-compile
Wall time (s)
H2 fixed
H2 runtime-coeff.
H3+ fixed
H3+ runtime-coeff.
D one compile + 1000 evaluations
61.7x
7.0x
82.2x
16.9x
no Catalyst
Catalyst compile-once
Modeled wall time (s)